

Sampler.Like.Audio Architectural Brief

AES60 Core Audio Metadata & Sampler.Like.Audio (APK) Integration Summary

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1. Executive Overview

This document synthesizes the **AES60 Core Audio Metadata** standard and its broader ecosystem of AES technical metadata standards (including AES31, AES41, AES-X098C, iXML, and AES-X230), establishing an architectural mapping for the **Sampler.Like.Audio (APK)** software sampler engine and sample library structure.

By harmonizing classic hardware sampler workflows (such as pad banks, choke/mute groups, keygroup program chromamapping, velocity switching, and zero-crossing chop points) with formal standards-compliant audio object schemas, Sampler.Like.Audio ensures lossless session transfer, non-destructive editing provenance, and archival integrity across modern production ecosystems.

2. Core Metadata Standards Matrix

AES60-2011 & EBU Tech 3293

Core Audio Metadata framework defining Dublin Core based metadata elements for broadcast and studio audio objects (title, creator, format, channel assignment, rights).

AES31 (Parts 1 - 4)

Digital audio file interchange, FAT32 disk formatting, Broadcast Wave Format (BWF), Simple Project Exchange (ADL), and XML-based Audio Decision Lists.

AES41 (Parts 1 - 5)

In-band embedded metadata within PCM audio sample LSBs for recoding decision data, MPEG Layer II, AAC, Dolby E, and EBU loudness/true-peak parameters.

AES-X098C & iXML

Administrative & structural processing history for audio objects (signal graph, hardware chain, DSP transformations), combined with location recording track/take metadata.

3. Sampler.Like.Audio (APK) Architecture & AES Metadata Mapping

The **Sampler.Like.Audio (APK)** engine structures digital sample playback, pad performance, and sample library management into standardized metadata nodes:

Pad & Voice Engine Mapping

- **Pad & Program Objects (APK 404 - 484):** Each pad kit (e.g., Kick, Snare, Closed Hat, Open Hat, Toms) is assigned an AES60 `audioTrack` and `channelIndex` , maintaining explicit track function metadata.
- **Velocity Layering (Up to 4 Layers per Pad):** Multi-velocity sample layers are mapped to AES60 technical metadata parameters, enabling dynamic response curves.
- **Mute / Choke Groups:** Explicit note-off routing triggers defined in AES31 ADL event decision structures to handle real-time voice choking (e.g., Open Hi-Hat cut by Closed Hi-Hat).
- **Non-Destructive Slice Points:** Chop mode slices (start, end, loop, zero-crossing points) are recorded as AES31 XML ADL region events without modifying underlying `.wav` sample files.

Preservation & Processing History (AES-X098C)

Every audio processing action performed inside the editor (bit-crushing, 12-bit decimation, analog filter modeling, ADSR envelopes, pitch-shifting) is tracked as a `processEvent` within an AES-X098C compliant XML signal graph record.

4. Complete Directory Inventory (28 AES Standards Files)

The table below catalogues all 28 specification documents and reference materials residing in `Documentation/AES Standards - Lenses/AES60/AES 60 - META DATA/` :

Filename	Format	Size	Standard Title & Functional Scope
<code>aes60-2011-f.pdf</code>	PDF	778 KB	AES60-2011: Core Audio Metadata Standard (EBU Core base)
<code>AES60_EBUtech3293v1_8.pdf</code>	PDF	1.27 MB	EBU Tech 3293 v1.8: EBUCore Specification (Foundation for AES60)
<code>ExtendAES60.pdf</code>	PDF	183 KB	AES60 - Extending EBU Core XML Schemas (NRK Technical Guidelines)

Filename	Format	Size	Standard Title & Functional Scope
aes31-1-2001-s2012-f.pdf	PDF	465 KB	AES31-1: Network & File Transfer — Part 1: Disk Format
aes31-2-2019-f.pdf	PDF	429 KB	AES31-2: Broadcast Wave Format (BWF) File Interchange
aes31-3-2008-r2019-f.pdf	PDF	611 KB	AES31-3: Simple Project Exchange (Edit Decision List / ADL)
aes31-4-2015-f.pdf	PDF	486 KB	AES31-4: XML Implementation of Audio Decision Lists (ADL XML)
aes31-4-151112-pcfc-sec.pdf	PDF	549 KB	Draft AES31-4: Committee Draft for ADL XML Revision
aes41-1-2012-r2017-f.pdf	PDF	522 KB	AES41-1: Audio-Embedded Metadata — Part 1: Architecture
aes41-2-2012-r2017-f.pdf	PDF	835 KB	AES41-2: MPEG-1 Layer II / MPEG-2 LSF Recoding Data Set
aes41-3-2012-r2017-f.pdf	PDF	544 KB	AES41-3: AAC & HE-AAC Embedded Metadata
aes41-4-2012-r2017-f.pdf	PDF	457 KB	AES41-4: Dolby E Embedded Metadata
aes41-5-2012-r2017-f.pdf	PDF	459 KB	AES41-5: EBU Loudness, True-Peak, and Downmix Metadata
AES41-r-AJM-20110325.pdf	PDF	237 KB	AES41 Embedded Metadata Working Paper by A. J. Mason
aes-r09-2008-f.pdf	PDF	329 KB	AES-R9-2008 Report: Considerations for Standardising Metadata Sets
iXML_schema_printv2b_120428.pdf	PDF	99 KB	iXML Schema Specification (Location Sound Metadata Payload)

Filename	Format	Size	Standard Title & Functional Scope
pif-x220-revised-140501.docx	DOCX	245 KB	PIF AES-X220: Professional Audio Definition Model Review
pif-x220-revised-140501(1).docx	DOCX	245 KB	PIF AES-X220: Duplicate Copy
x98c10112014.docx	DOCX	475 KB	AES-X098C Draft Standard: Processing and Handling History
x98c10112014(1).docx	DOCX	475 KB	AES-X098C Draft Standard: Duplicate Copy
AES-x98c_2011-12-05MS.docx	DOCX	291 KB	AES-X098C Manuscript Draft Template
AES-x98c_2011-12-05MS(1).docx	DOCX	291 KB	AES-X098C Manuscript Draft Template: Duplicate Copy
X230-Mic-Profile-ObjectList-180502(1).xlsm.xlsx	XLSX	66 KB	AES-X230 Microphone Profile Object List Spreadsheet
X230-Mic-Profile-ObjectList-180502(2).xlsm.xlsx	XLSX	66 KB	AES-X230 Microphone Profile Object List: Duplicate Copy
18830.pdf	PDF	21.87 MB	JAES Vol. 12 No. 2 (1964) Journal Issue (Dynagroove, Tape Design)
Architecture_for_Blockchain.pdf	PDF	734 KB	Cloud Customer Architecture for Blockchain (CSCC / OMG Overview)
CREATe-Blockchain-2016-05.pdf	PDF	1.36 MB	CREATe Working Paper: Blockchain in the Music Industry
MS_619133_CBC_Telex.pdf	PDF	280 KB	Applied Electronics Commercial Quotation to CBC/Radio-Canada

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